

LEARNER GUIDE

Unit Standard Title: PERFORM BASIC FIRE FIGHTING
Unit Standard No: 12484
Unit Standard Credits: 4
NQF Level: 2

In terms of the Copyright Act 1978:

No part of this manual may be reproduced or translated in any form or by any means, electronic or mechanical including photocopying, recording or by an information storage and retrieval system without the express permission from TDA.

COPYRIGHT: TRAINING DYNAMICS AFRICA

First edition 2008

Publications disclaimer

Neither the authors nor the SETA can accept any responsibility for any legal or any other consequences, which may arise directly or indirectly as a result of the use or adaptation of or reliance upon any of the content of this material.

This material has been compiled based on the following unit standard:

12484: Perform basic fire fighting

All the above learning material has been accredited by SASSETA in the Republic of South Africa.

Material compiled by:

Training Dynamics Africa

Learning program guide	5-9
1. Introduction	5
2. Purpose of this learning program	5
3. Standards and qualifications	5
4. Assessments	6
5. Range statements	6
6. The learning guide	6
7. Learning assumed to be in place	7
8. Preparation of summative assessment	9
9. Roles and responsibilities	9
Study Unit 1: Types of Fire	11-17
1. What is a fire	13
2. Elements of a fire	13
3. There are four types of fires	14
4. What causes a fire	15
Study Unit 2: Fire prevention	19-27
1. Goals of fire prevention	21
2. Strategy to prevent a fire.	22
3. General measures to prevent a fire.	23
Study Unit 3: Housekeeping	29-35
1. Why is good housekeeping important?	31
2. General housekeeping guidelines	31
3. Storage guidelines	33
4. DO'S and DON'T'S	34
Study Unit 4: Fire fighting equipment	37-50
1. Types of fire extinguishers	39
2. Identification of fire extinguishers	41
3. How does a fire extinguisher work	4
4. How to operate a fire extinguisher	46
5. Fire hoses	49
Study Unit 5: Fighting Fires	51-58
1. Fighting a fire	53
2. Stopping a fire	55

3. Precaution when fighting a fire.	56
4. Tips for safe extinguisher use:	56
5. Procedure after the fire has been put out.	57

Study Unit 6: Maintenance	59-63
----------------------------------	--------------

1. Introduction	61
2. Don't fight a fire unless.	61

Unit Standard: 12484 Perform basic fire fighting.	65
--	-----------

1. Introduction

This learning program is part of a complete qualification. For more details concerning the complete qualification please contact your service provider or alternatively the office of the SASSETA

2. Purpose of this learning program

The skills, values and knowledge reflected in this unit standard are required by people in the field of manufacturing and engineering. The learning outcomes in this unit standard also contribute to the exit level outcomes required for various manufacturing and engineering qualifications.

Qualifying learners can select and use the appropriate fire fighting equipment to extinguish or control fires in the workplace.

2.1 Target group

This program is compiled for the following **target group**:

- Security members
- South African Defense Force members
- South African Police Force members
- Correctional Services
- Individuals who wishes to complete the NQF level 3 National Certificate in security practices.

3. Standards and qualifications

It is important that you as the learner understand and realize that this training program is not presented in isolation, but that it was developed based upon nationally recognized standards known as unit standards.

Unit standards are the “building blocks” of qualifications. All qualifications are plotted on the National Qualifications Framework (NQF).

REMEMBER.....

Unit standards comprises of outcomes. An outcome is a statement that describes the required competency that must be demonstrated by the learner on successful completion of a training intervention.

4. Assessments

In order to assess whether a learner can actually demonstrate the desired outcomes, assessment criteria are included in the unit standard. Each outcome has its own set of assessment criteria.

REMEMBER.....

The assessment criteria describe the evidence that is needed that will show that the learner has demonstrated the outcome correctly.

Kindly refer to the unit standard attached hereto for the assessment criteria.

It is of utmost importance that the learner fully understands the assessment criteria as listed in the unit standard, as it is the only way in which the learner will know what he will be assessed against.

The **final or summative assessment** is the most important aspect of this training program. It is during this process that the learner will be declared competent or not yet competent.

The learner will know exactly how he will be assessed, and when and where he will be assessed. All of these details must be obtained from the training provider where the learner enrolled for his program.

5. Range statements

REMEMBER.....

Also included in the unit standard are the range statements in support of the assessment criteria.

The range statements indicate detailed requirements of the assessment criteria.

6. The learning guide

The learner guide is included in this material under various learning units. The learner guide has been designed in such a manner that the learner is guided in a logical way through the learning material and requirements of the unit standard.

On completion of this skill program, the learner will be assessed against the assessment criteria as stipulated in the unit standard.

Each of the modules consists of learner outcomes to be achieved during training and development interventions, work related experience or at your own time. The learner outcomes will enable you to achieve the required level of confidence and competence in order to undergo the summative (final evaluations) assessment.

The learner needs to assess his own knowledge and skill throughout the training process by completing the learner workbook. With the completion of the workbook you start collecting evidence to proof your competence. The learner workbook will be assessed (FORMATIVE ASSESSMENT)

TAKE NOTE

The learner guide will remain the property of the learner once the LEARNING PROGRAM has been completed.

TAKE NOTE

- The best results will be obtained if you start with Study Unit 1 in the Learner Guide, and work your way through it, study unit by study unit.
- All learning outcomes are vital and must be studied (and exercised for all practical requirements) thoroughly to ensure that enough evidence for all specific outcomes is generated.

7. Learning assumed to be in place

It is assumed that a learner which registers for this program have completed a program in communication at NQF Level 3 or equivalent.

It is imperative to understand that this program are only one of the building blocks to complete Skills program 1 The complete list of unit standards for Skills program appears on the next page.

7.1 Security program matrix

SKILLS PROGRAM 1: SASSETA E				Patrol Security officer/Grade E
244177	Conduct a security patrol in area of responsibility	Level 3	7 Credits	
244184	Apply legal aspects in a security environment	Level 3	8 Credits	
244176	Use security equipment	Level 2	2 Credits	
244179	Handle complaints and problems	Level 3	6 Credits	
244181	Perform hand over and take over responsibilities	Level 3	2 Credits	
244182	Give evidence in court	Level 3	4 Credits	
12484	Perform basic fire fighting	Level 2	4 Credits	
116534	Carry out basic first aid treatment in the work-place	Level 3	2 Credits	
SKILLS PROGRAM 2: SASSETA D				Access control officer/Grade D
244189	Conduct access and egress control	Level 4	7 Credits	
242825	Conduct evacuations and emergency drills	Level 4	4 Credits	
11505	Identify, handle and defuse security related conflict	Level 4	12 Credits	
117705	Demonstrate knowledge of the Firearms Control Act 2000 (Act No. 60 of 2000)	Level 3	3 Credits	
113924	Apply basic business ethics in a work environment	Level 2	2 Credits	
119465	Write/present/sign texts for a range of communicative contexts	Level 3	5 Credits	
114979	Operate a computer workstation in a business environment	Level 3	2 Credits	
SKILLS PROGRAM 3: SASSETA C				Asset & reaction officer/Grade C
113909	Coach a team member in order to enhance individual performance in work environment	Level 3	5 Credits	
13912	Apply knowledge of self and team in order to develop a plan to enhance team performance	Level 3	5 Credits	
244578	Describe how to manage reactions arising from a traumatic event	Level 3	2 Credits	
113852	Apply occupational health, safety and environmental principles	Level 3	10 Credits	
13936	Outline the legal environment of a selected industry	Level 3	2 Credits	
119472	Accommodate audience and context needs in oral/signed communication	Level 3	5 Credits	
11983	Compile and administer a case docket for investigation purposes	Level 5	6 Credits	

UNIT STANDARD 12484: Perform basic fire fighting

8. Preparation of summative assessment

After you have worked through the **LEARNER GUIDE** and you are satisfied that all theoretical and practical requirements/ evidence, as stipulated by the UNIT STANDARD, can be met, you need to make your own necessary arrangements with the assessor and/or facilitator. The training establishment will then arrange a suitable date for you to attend your pre-assessment meeting and then arrange for a suitable date to complete the final assessment.

8.1 RPL assessment



The assessment of RPL learners will be conducted in the same way as for those of new learners. The assessment pack is exactly the same and will therefore be used for new learners as well as RPL Learners. It must however be noted that learners who are applying for RPL must provide proof of previous learning and subject related experience prior to the assessment.

This proof or evidence can be in the format of certified copies (certificates) of previous learning programs that have been attended.

All the evidence will be assessed and authenticated before a learner will be allowed to enroll for an RPL program.

9. Roles and responsibilities

9.1 Facilitator/instructor

- Prepare to facilitate.
- Transfer of knowledge and skills

9.2 The role of the assessor

- Plan for the assessment
- Prepare the learner for the assessment

9.3 The role of the Learner

- Prepare for assessment.
- Accept the responsibility of learning.
- Accept the responsibility to produce/deliver evidence.

STUDY UNIT 1

TYPES OF FIRES

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Define a fire.
- ♦ Identify the elements of a fire.
- ♦ Demonstrate knowledge of the different types of fire.
- ♦ Identify the various causes of a fire.

Define a fire.



1. What is a fire?



It's a visible, tangible side effect of matter changing form: its one part of a chemical reaction.

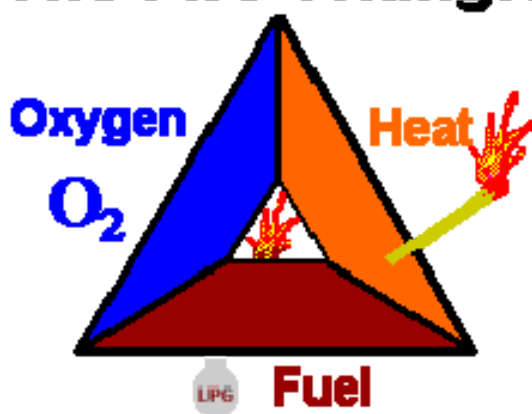
Identify the elements of a fire.



2. Elements of a fire.

For fires to exist, the following four elements must be present at the same time:

The Fire Triangle



- ♦ Enough **oxygen** to sustain combustion,
- ♦ Enough **heat** to raise the material to its ignition temperature,
- ♦ Some sort of **fuel** or combustible material, and
- ♦ The **chemical reaction** that is fire.

Demonstrate knowledge of the different types of fire.



3. There are four types of fire

CLASSES OF FIRE		DESCRIPTION
A	 Common materials, such as paper, wood or most other combustibles.	Paper/trash/wood and other more-or-less organic solids. "Ordinary combustibles."
B	 Flammable liquids, such as gasoline paint remover or grease.	Inflammable liquids such as gasoline or paint thinner.
C	 Electrical fires.	Electrical fires, with electricity still flowing to the burning equipment.
D	 Combustible metals, usually found in industry.	Burning reactive metals, such as sodium, magnesium, titanium, and so forth. Such metals not only burn at high temperatures but can chemically extract oxygen from water and even carbon dioxide.

Identify the various causes of a fire.



4. What causes a fire

The following issues are only some the reasons why fires occur:

4.1 Careless smoking



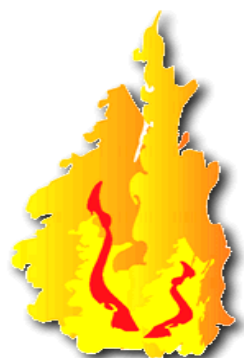
Careless smoking is the leading cause of residential fire deaths. Smoke detectors and smolder-resistant bedding and upholstered furniture are significant fire deterrents.

4.2 Arson



Arson is the second largest cause of residential fires and residential fire deaths. In commercial properties, arson is the major cause of deaths, injuries and dollar loss.

4.3 Heating



Heating is the third leading cause of residential fire deaths. Heater fires are the leading cause of fire deaths.

4.4 Cooking



Cooking is the leading cause of apartment fires and the second most frequent cause of single-family residential fires. These fires often result from unattended cooking and human error, rather than from mechanical failures of stoves or ovens.

4.5 Wiring,



Wiring, outlets, switches, circuit breakers and other electrical devices are the third leading cause of fires and the second leading cause of fire deaths.

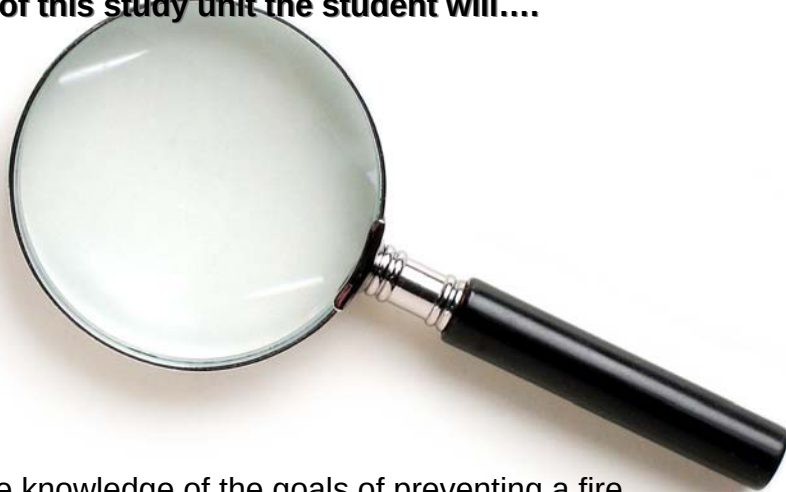
[illegible]

STUDY UNIT 2

FIRE PREVENTION

LEARNING OUTCOMES

On completion of this study unit the student will....

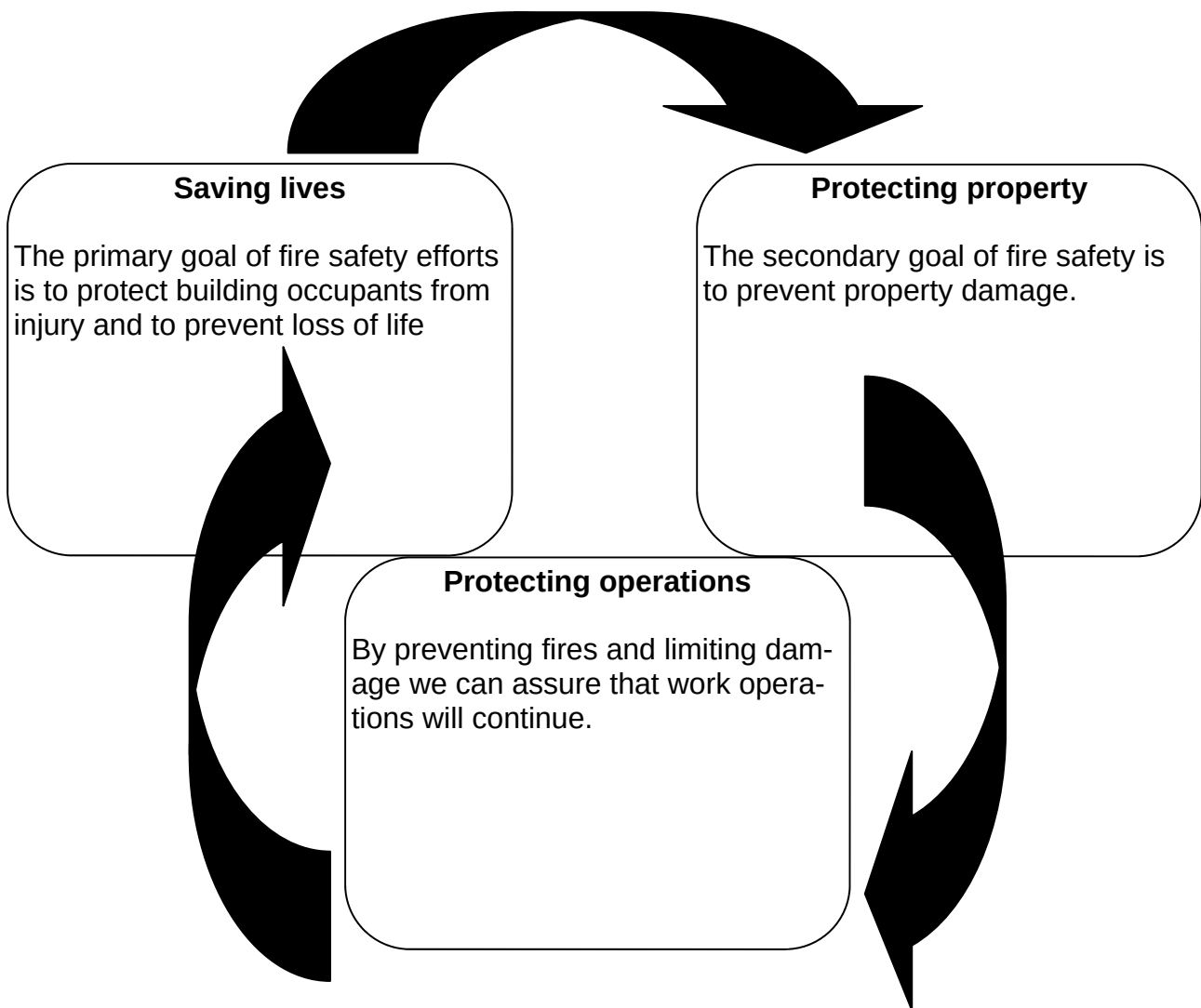


- ♦ Demonstrate knowledge of the goals of preventing a fire.
- ♦ Demonstrate skills and knowledge to setup a strategy to prevent a fire.
- ♦ List general measures to prevent a fire.

Demonstrate knowledge of the goals of preventing a fire.



1. GOALS OF FIRE PREVENTION.



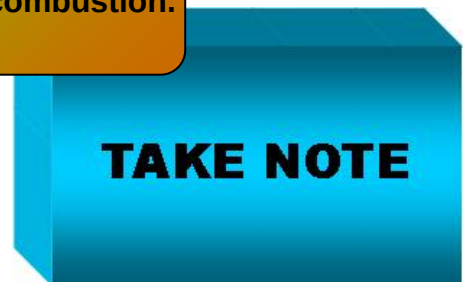
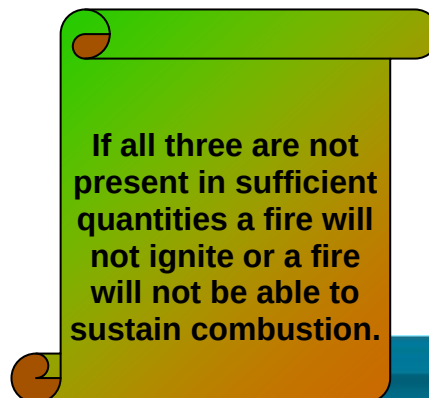
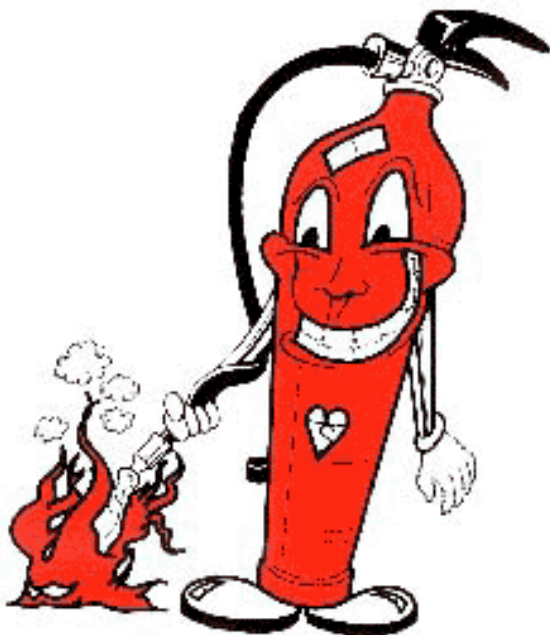
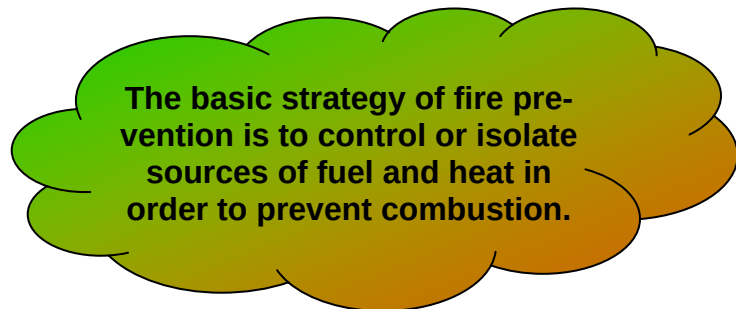
Demonstrate skills and knowledge to setup a strategy to prevent a fire.



2. STRATEGY TO PREVENT A FIRE.

A fire must have three things to ignite and maintain combustion:

- ♦ Fuel
- ♦ Heat
- ♦ Oxygen



List general measures to prevent a fire.



3. GENERAL MEASURES TO PREVENT A FIRE.

- ♦ **Smoke Detector:** It is important to have smoke detectors installed on each floor.

SAFETY FIRST!



Remember to check them regularly for proper function and change their battery at least once a year.

REMEMBER.....



- ♦ **Emergency Numbers:** While you should call emergency services if there is a fire, do not call from the location of the fire. You will be wasting valuable time you can be using to save your life.



Remember to update your emergency number list regularly.

REMEMBER.....

- ♦ **Have a plan:** Make sure everyone is familiar with the evacuation plan and the location of the emergency doors. This is because if one exit route is blocked, you have a backup plan. Also, agree on a place to meet to do a “head count” once you evacuate the building.

HAVE A PLAN

REMEMBER.....

Always clear the emergency doors and exit routes from unnecessary objects.

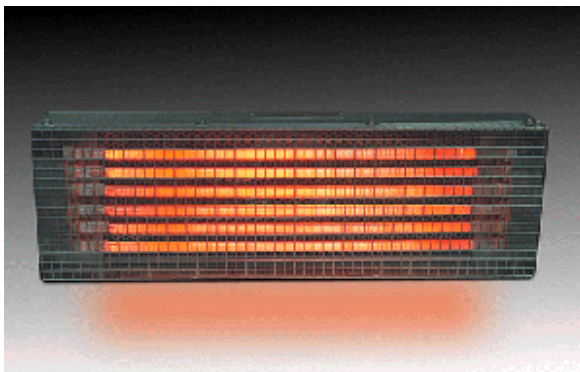
- ♦ **Smokers:** Never smoke while going to sleep or when you are sleepy.



REMEMBER.....

The ashes could easily start a fire while you are too drowsy to be fully alert.

- ♦ **Heaters:** Any portable heater should be kept at least a meter from anything that can burn (including the wall).



REMEMBER.....

Never leave a heater on when you are not in the office.

- ♦ **Matches:** It is very important to **keep matches and lighters out of a child's reach**. Also, do not keep any near your fireplace.



REMEMBER.....

Half of all fire deaths occur at night, so fire hazard checks and special attention to fire prevention should occur before applying your mind on something else.

- ♦ **Cooking:** Keep the cooking areas clear of combustibles. Make sure to keep all pot handles turned inward so they are not accidentally knocked over. Try to wear short sleeves or fitted clothes so your sleeves don't catch fire.



REMEMBER.....

In case a grease fire does occur, carefully and calmly slide a lid over the pot to smother the flames and turn off the burner.

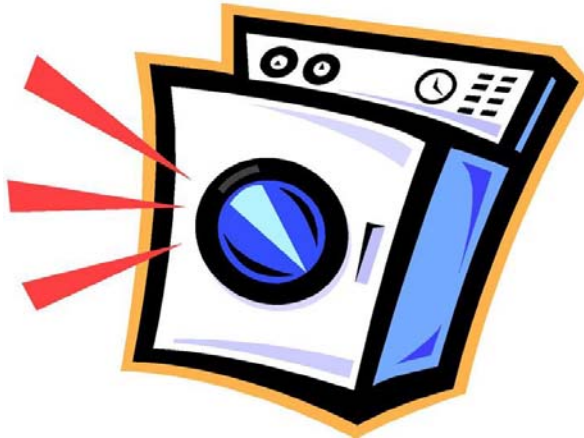
- ♦ **Fireplace:** Before each new heating season, it is a good idea to have your chimney cleaned. Also, do not keep any matches, newspapers, or kindling near the fireplace.



REMEMBER.....

Use a fireplace screen to prevent any sparks from flying out.

- ♦ **Clothes Dryer:** Never leave your clothes dryer on when you are not home.



- ♦ **Electric appliances & wiring**

- ❖ Replace or repair loose or frayed cords on all electrical devices.
- ❖ If outlets or switches feel warm, shut off the circuit and have them checked by an electrician.
- ❖ Try to avoid extension cords. If you feel an extension cord is necessary, make sure that it is not frayed or worn. Do not run it under carpet or around doorways.
- ❖ If a circuit breaker trips or a fuse blows frequently, cut down on the number of appliances on that line.
- ❖ Assure there's plenty of airspace round entertainment units such as the TV and stereo to avoid overheating.



[illegible]

STUDY UNIT 3

HOUSEKEEPING

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Demonstrate the importance of house keeping
- ♦ Apply good housekeeping guidelines according to company SOP.
- ♦ Demonstrate knowledge of storage guidelines.

Introduction

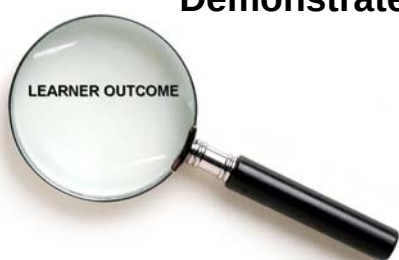
Good housekeeping habits are an important part of a safe workplace.

Effective house keeping can eliminate some workplace hazards and help get a job done easily and properly.

A good housekeeping program plans and manages the orderly storage and movement of materials from point of entry to exit. It includes a material flow plan to ensure minimal handling.

Employees should be reporting any unusual conditions or hazards as well as obeying posted warning signs.

Demonstrate the importance of house keeping.



1. Why is good housekeeping important?



- ◆ To reduce amounts of flammable and combustible materials.
- ◆ To reduce ignition hazards.
- ◆ To ensure safe emergency evacuation of occupants.
- ◆ To allow for quick emergency response.

Apply good housekeeping guidelines according to company SOP.



2. GENERAL HOUSEKEEPING GUIDELINES.

- ❖ Work areas, aisles, walkways, stairways, and equipment should be kept clear of loose materials, trash, scraps, etc.



Inspection is the only way to check for deficiencies in the workplace. The importance hereof is critical for the necessary changes to be made.

TAKE NOTE

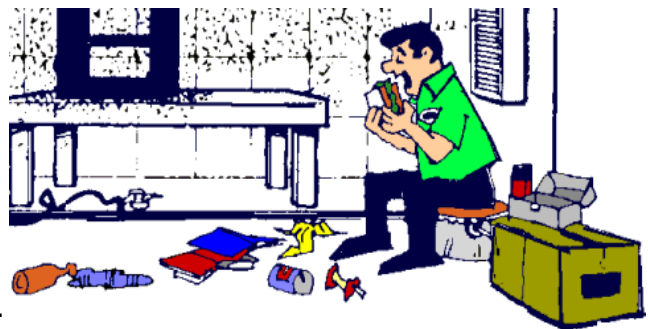
- ◆ Never block aisles, fire exits, emergency equipment, or alarm pull stations with equipment or materials.



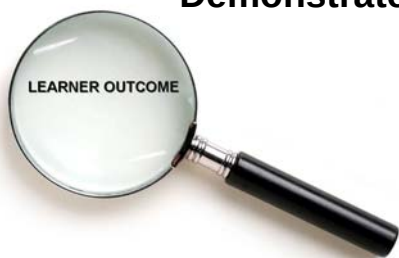
- ❖ Avoid build up of combustible trash and waste such as paper, wood, cardboard, etc.



- ❖ Keep use and storage of flammables and combustibles to a minimum.



- ❖ Clean up all spills such as grease, oil, or water immediately. A delay could result in accidents.

Demonstrate knowledge of storage guidelines.**3. STORAGE GUIDELINES.**

- ❖ No storage is allowed in corridors and stairwells. A cluttered hallway could slow down emergency evacuation.



- ❖ Storage must not exceed plane of 18 inches below sprinkler heads or smoke detectors. Storage that breaks this plane may prevent sprinkler heads from fully covering room during a fire.
- ❖ All storage must be at least 3 ft from electrical panels. In some emergency situations it will be necessary to access these panels quickly.

REMEMBER.....

Flammable and combustible liquids are potential fuel sources for fires and are present in almost every workplace.

IMPORTANT**TAKE NOTE**

It is actually the vapor created by flammable and combustible liquids that ignite and burns.

It is important to understand what materials in your work area are flammable and combustible so that you may properly store and isolate them from ignition sources.

4. DO'S & DON'T'S.

DO'S	DON'T'S
Keep floors clean and clear of waste.	Use aisles and stairways as storage areas.
Keep workplaces adequately lighted.	Allow materials to build up on floors.
Keep light fixtures clean.	Block emergency exits, fire equipment or first aid stations with stored materials.
Inspect and clean machinery and tools regularly.	Store compressed gases near heat sources.
Clean up spills immediately.	

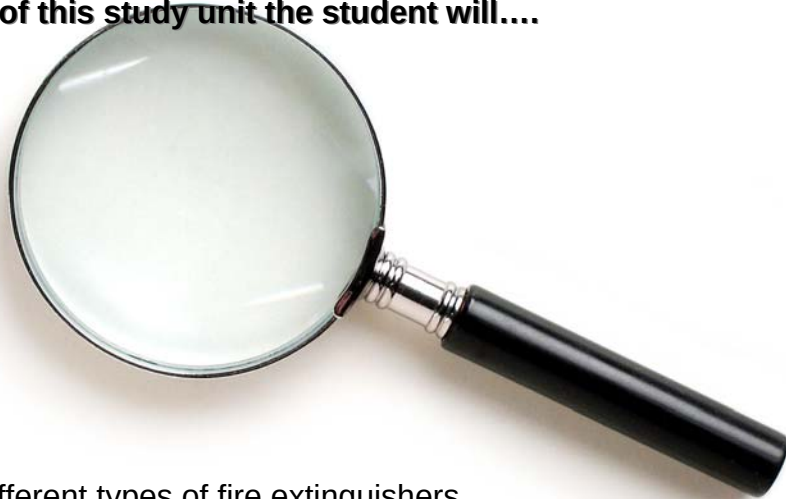
This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

STUDY UNIT 4

FIRE FIGHTING EQUIPMENT

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Know the different types of fire extinguishers.
- ♦ Know the use of the different types of fire extinguishers.
- ♦ Identify the correct fire extinguisher to fight a specific class of fire..
- ♦ How to operate a fire extinguisher..
- ♦ Know the location and operation of a fire hose.

Know the different types of fire extinguishers.



Know the use of the different types of fire extinguishers.

1. Types of fire extinguishers

Fire extinguishers are divided into four categories, based on different types of fires. Each fire extinguisher also has a numerical rating that serves as a guide for the amount of fire the extinguisher can handle.

TAKE NOTE

The higher the number,
the more fire-fighting
power.



The following is a quick guide to help choose the right type of extinguisher.

EXAMPLE	TYPE
	Class A: Extinguishers are for ordinary combustible materials such as paper, wood, cardboard, and most plastics. The numerical rating on these types of extinguishers indicates the amount of water it holds and the amount of fire it can extinguish.
	Class B: Fires involve flammable or combustible liquids such as gasoline, kerosene, grease and oil. The numerical rating for Class B extinguishers indicates the approximate number of square feet of fire it can extinguish.
 REMEMBER..... Never use water to extinguish class C fires - the risk of electrical shock is far too great!	Class C: Fires involve electrical equipment, such as appliances, wiring, circuit breakers and outlets. Class C extinguishers do not have a numerical rating. The C classification means the extinguishing agent is non-conductive.
 REMEMBER..... they are designed for class D fires only.	Class D: Fire extinguishers are commonly found in a chemical laboratory. They are for fires that involve combustible metals, such as magnesium, titanium, potassium and sodium. These types of extinguishers also have no numerical rating, nor are they given a multi-purpose rating

Identify the correct fire extinguisher to fight a specific class of fire.

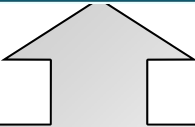


2. Identification of fire extinguishers

- ❖ Some fires may involve a combination of these classifications.
Your fire extinguishers should have ABC ratings on them.



IDENTIFICATION	TYPE
  	Water extinguishers or APW extinguishers (air-pressurized water) are suitable for Class A fires ONLY. Water extinguishers are filled with water and pressurized with oxygen.  Water extinguishers can be very dangerous in the wrong type of situation.  Only fight the fire if you're certain it contains ordinary combustible materials only.

IDENTIFICATION	TYPE
  Never use a water extinguisher on grease fires, electrical fires or class D fires. The flames will spread and make the fire bigger! 	Water extinguishers or APW extinguishers (air-pressurized water) are suitable for Class A fires ONLY. Water extinguishers are filled with water and pressurized with oxygen.  Water extinguishers can be very dangerous in the wrong type of situation.  Only fight the fire if you're certain it contains ordinary combustible materials only.



Dry chemical extinguishers come in a variety of types and are suitable for a combination of **Class A, B and C fires**.

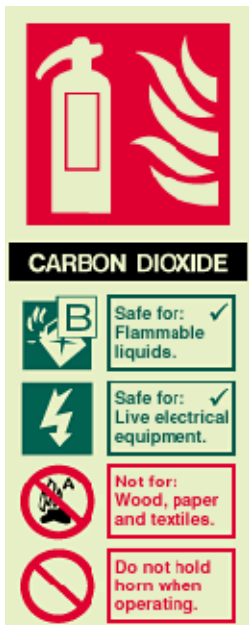
These are filled with foam or powder and pressurized with nitrogen.

BC: This is the regular type of dry chemical extinguisher. It is filled with sodium bicarbonate or potassium bicarbonate. The BC variety leaves a mildly corrosive residue which must be cleaned immediately to prevent any damage to materials.

ABC: This is the multipurpose dry chemical extinguisher. The ABC type is filled with mono-ammonium phosphate, a yellow powder that leaves a sticky residue that may be damaging to electrical appliances such as a computer.

REMEMBER

They don't work very well on class A fires because they may not be able to displace enough oxygen to put the fire out, causing it to re-ignite.



Carbon Dioxide (CO₂) extinguishers are used for **Class B and C fires**.

CO₂ extinguishers contain carbon dioxide, a non-flammable gas, and are highly pressurized. The pressure is so great that it is not uncommon for bits of dry ice to shoot out the nozzle.

CO₂ extinguishers have an advantage over dry chemical extinguishers since they don't leave a harmful residue, a good choice for an electrical fire on a computer or other favorite electronic device such as a stereo or TV.

CRITICAL

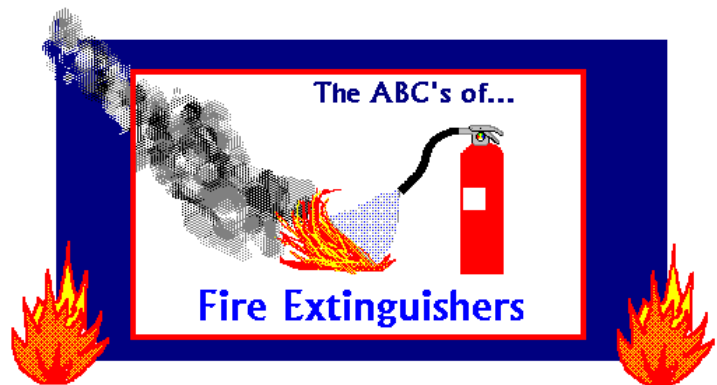
It is vital to know what type of extinguisher you are using. Using the wrong type of extinguisher for the wrong type of fire can be life-threatening.

How to operate a fire extinguisher



3. How does a fire extinguisher work?

Fire needs fuel, oxygen and heat in order to burn. In simple terms, fire extinguishers remove one of these elements by applying an agent that either cools the burning fuel, or removes or displaces the surrounding oxygen.



Fire extinguishers are filled with water or a smothering material, such as CO₂. By pulling out the safety pin and depressing the lever at the top of the cylinder (the body of the extinguisher), this material is released by high amounts of pressure.

Although the temptation is to aim the extinguisher at the flames, the proper way to use the extinguisher is to aim it directly at the fuel.



3.1 Water Extinguishers



Water extinguishers are filled with regular tap water and pressurized with oxygen. The best way to remove heat is to dump water on the fire but, depending on the type of fire, this is not always the best option.

3.2 Dry Chemical Extinguishers

Dry chemical extinguishers are filled with either foam or powder, usually sodium bicarbonate (baking soda) or potassium bicarbonate, and pressurized with nitrogen. Baking soda is effective because it decomposes at 158 degrees Fahrenheit and releases carbon dioxide (which smothers oxygen) once it decomposes.



Dry chemical extinguishers interrupt the chemical reaction of the fire by coating the fuel with a thin layer of powder or foam, separating the fuel from the surrounding oxygen.

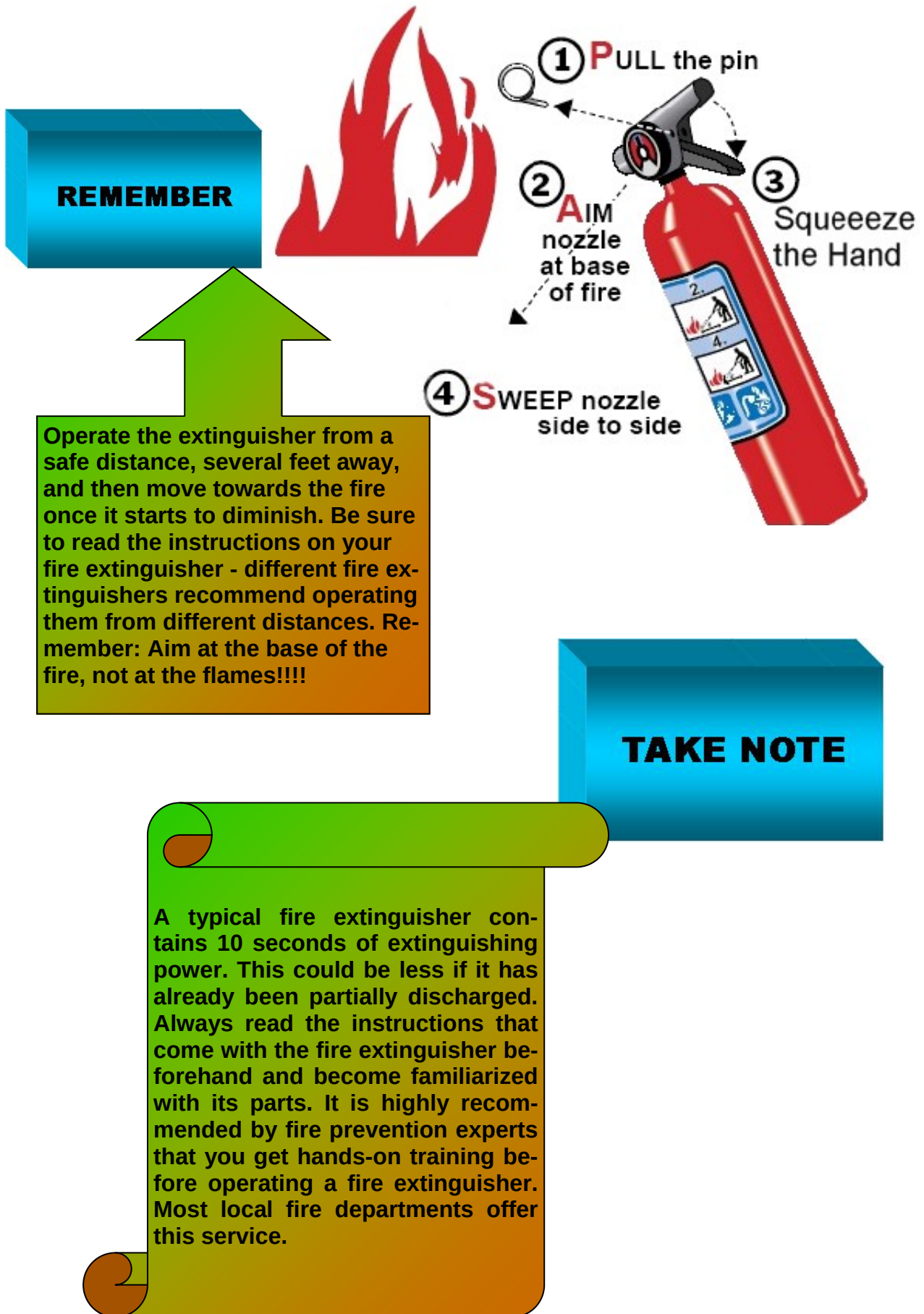
3.3 Carbon Dioxide (CO2) extinguishers



CO2 extinguishers contain carbon dioxide, a non-flammable gas, and are highly pressurized. The pressure is so great that it is not uncommon for bits of dry ice to shoot out. CO2 is heavier than oxygen so these extinguishers work by displacing or taking away oxygen from the surrounding area. CO2 is also very cold so it also works by cooling the fuel.

4. HOW TO OPERATE A FIRE EXTINGUISHER





4.1 The PASS method

The following method is the most commonly known operation of a fire extinguisher:

"P" stands for **PULL** the pin.



This will unlock the operating handle and allow you to discharge the extinguisher.

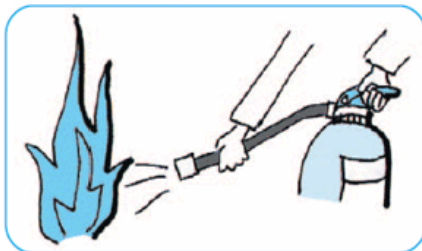
"A" stands for **AIM**



.....at the base of the fire.

This is important - in order to put out the fire, you must extinguish the fuel.

"S" stands for **SQUEEZE**



This will release the extinguishing agent in the extinguisher. If the handle is released, the discharge will stop.

"S" stands for **SWEEP**



Using a sweeping motion, move the fire extinguisher back and forth until the fire is completely out.

Know the location and operation of a fire hose.



5. FIRE HOSES

- ❖ The large fire hose reels located in multi-story buildings are to be used by building occupants to fight fire, especially when they are trapped and cannot escape to an **EMERGENCY EXIT**.



- ❖ The fire hoses are connected to the mains water supply and extend for about 30 meter. Some fire hose reels are located in cabinets whilst others are visible on the wall in a hall or corridor. They will always have appropriate signage indicating their location.

5.1 How to operate a fire hose

- ❖ Fire hose reels are all very similar in operation. This is the generic procedure:
 - ◆ Ensure the nozzle or jet is in the closed position
 - ◆ Turn on the main valve (some will not let the nozzle out until this is done)
 - ◆ Pull the hose off the drum, towards the fire
 - ◆ Open the nozzle or valve and direct the stream of water at the fire



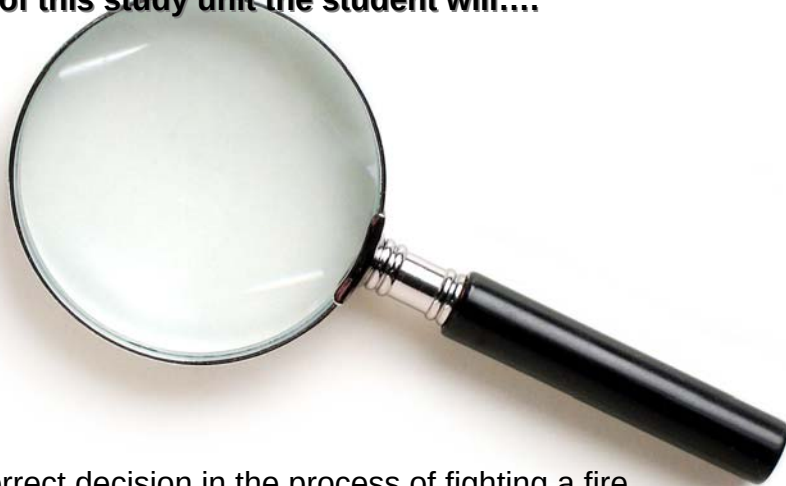
[illegible]

STUDY UNIT 5

FIGHTING FIRE

LEARNING OUTCOMES

On completion of this study unit the student will....



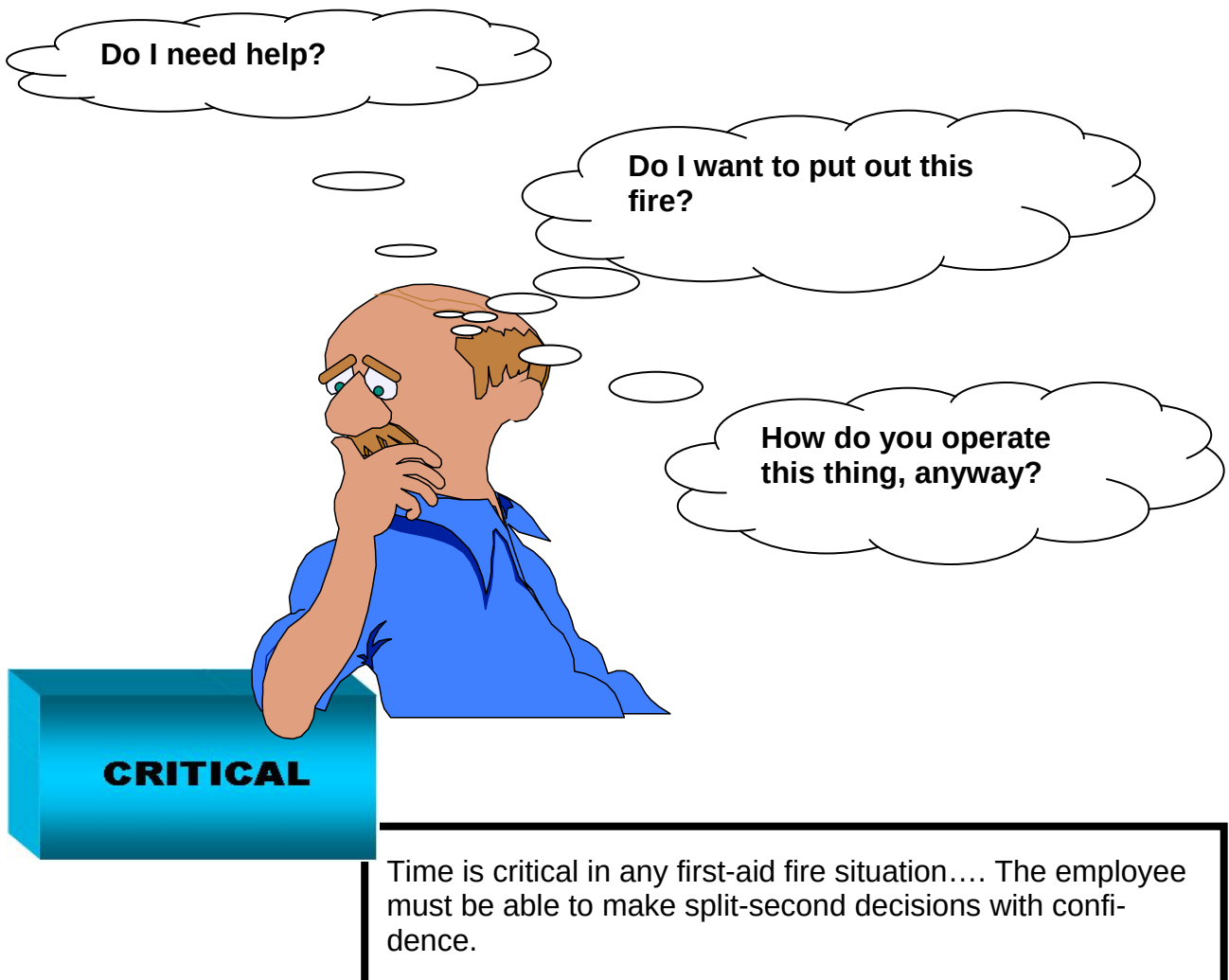
- ♦ Make the correct decision in the process of fighting a fire..
- ♦ Stop a fire by using the correct equipment and procedure..
- ♦ Apply general precautions when fighting a fire.
- ♦ Identify and apply the correct procedures after a fire has been put out accordingly to SOP.

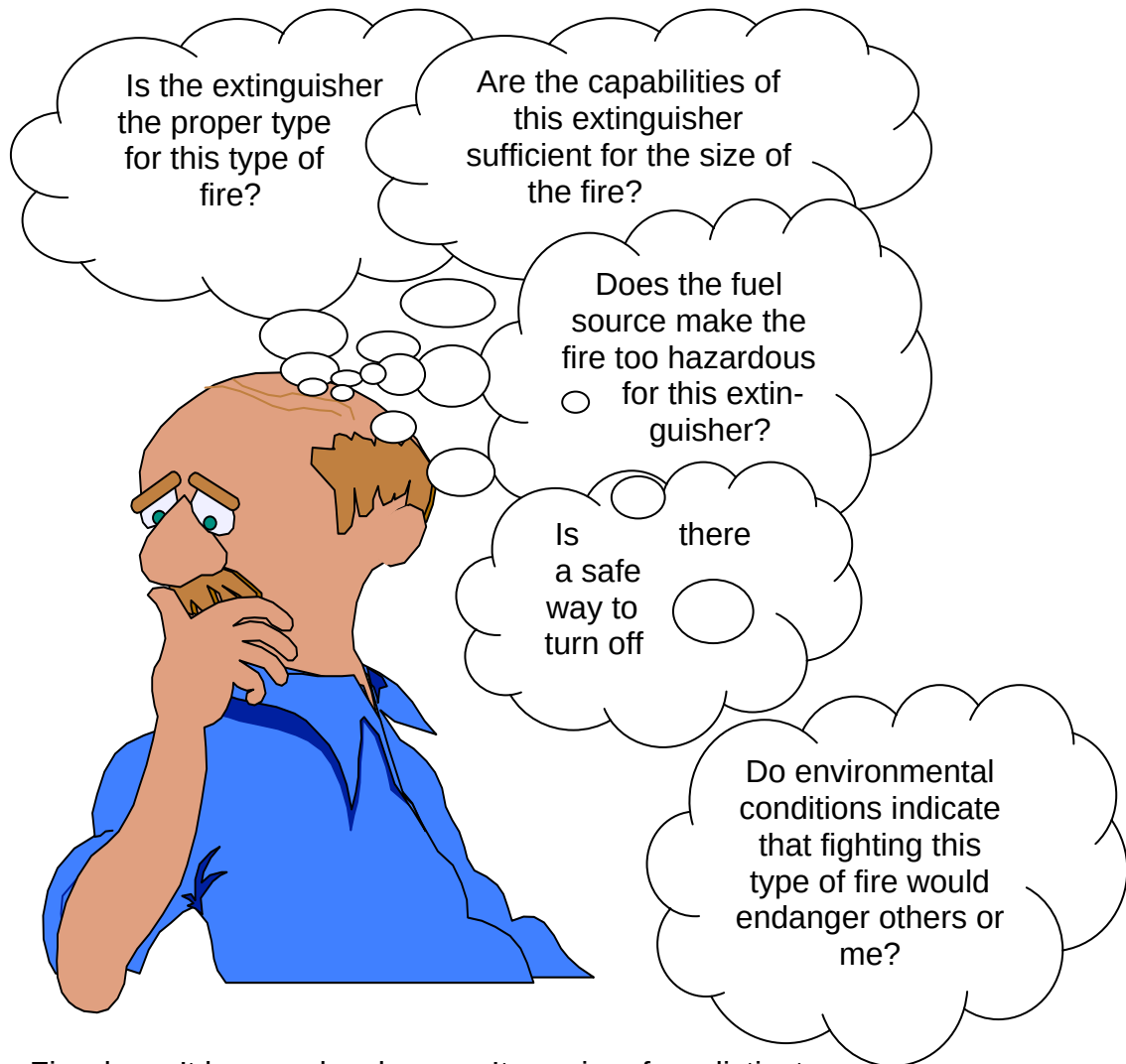


Make the correct decision in the process of fighting a fire

1. FIGHTING A FIRE

When a fire breaks out, you have only seconds to react, and in those few seconds, you must respond quickly and efficiently. You must stay calm and know your options. Frequently, the best option will simply be to leave the area and call for help. Yet, there may be other times when you will be in a position to put out the fire through the proper use of the right type of fire extinguisher.





- ♦ Fire doesn't happen by chance. It requires four distinct elements, and all four must be present in order for a fire to take place:

- ❖ Fuel (such as wood, paper, cloth, propane, gasoline, kerosene, Coleman fuel)
- ❖ Oxygen (16% of the air)
- ❖ Heat (for instance, a match or spark)
- ❖ The chemical reaction that results from fuel, oxygen and heat mixing in the right quantity, at the right time

If any one of the four elements (fuel, oxygen, heat or chemical reaction) is removed, a fire cannot survive. Fighting fire means eliminating one or more of these elements.

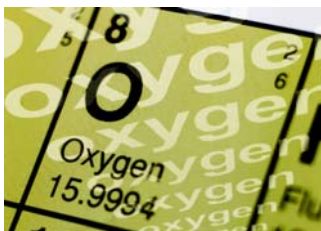
TAKE NOTE

Stop a fire by using the correct equipment and procedure.



2. STOPPING A FIRE

- ♦ Removing the fuel is difficult to do, since the fuel is what's actually burning. If the fuel consists of flammable vapors, special fire-fighting foams can occasionally control those vapors and in essence remove the fuel.



- ♦ Reducing or eliminating the fire's access to oxygen is much easier. This is referred to as "smothering" the fire.

- ♦ Cooling the fire is the method with which you are probably the most familiar. Spraying water on a fire cools the fuel long enough to stop the chain reaction.



2.1 General procedures when facing a fire.

Always keep the exit to your back while fighting a fire, and if your efforts don't seem to be successful, leave the area, closing doors behind you to assist in keeping the fire from spreading.

Once you are safe, call your local fire Department immediately.



A fire can be put out if you keep in mind a few simple details:

- ♦ Know how fires take place, the four elements of fuel, oxygen, heat and chain reaction.
- ♦ Know the different classes of fire.
- ♦ Know whether the fire extinguisher is the right class for the type of fire.
- ♦ Know how to correctly operate a fire extinguisher, using PASS: Pull, Aim, Squeeze, and Sweep.

Apply general precautions when fighting a fire.**3. PRECAUTION WHEN FIGHTING A FIRE**

The following general rule of precautions is important for the safety of ALL involved in a fire.

REACT *upon discovery* of fire or smoke

- ♦ **Remove** persons in immediate danger!
- ♦ **Ensure** doors are closed! (*confine fire/smoke*)
- ♦ **Activate** the building alarm!
- ♦ **Call** the Fire Department!
- ♦ **Treat** ALL fires as DANGEROUS!

**4. TIPS FOR SAFE EXTINGUISHER USE:**

Test that the extinguisher works before you approach the fire.

- ♦ Protect yourself at all times.
- ♦ Take care. Speed is essential but it is more important to be cautious.
- ♦ Keep your back to the exit at all times and stand 2 to 2.4m (6 to 8 ft.) away from the fire.

REMEMBER**The PASS word:**

- ♦ Get assistance **BEFORE** trying to fight a fire!

Identify and apply the correct procedures after a fire has been put out accordingly to SOP.



5. PROCEDURE AFTER THE FIRE HAS BEEN PUT OUT.

- ♦ WATCH the fire area
- ♦ If the fire flares up again repeat!
- ♦ If you can't control the fire, LEAVE immediately!
- ♦ Call the fire department to inspect the scene!
- ♦ Recharge or replace any used fire extinguisher!

[illegible]

STUDY UNIT 6

MAINTENANCE

LEARNING OUTCOMES

On completion of this study unit the student will....



- ♦ Apply basic maintenance on fire fighting equipment.

Apply basic maintenance on fire fighting equipment.



1. INTRODUCTION

Inspect fire extinguishers at least once a month (more often in severe environments). Fire extinguisher maintenance is important for everyone's safety.

You must ensure that:

- ◆ The extinguisher is not blocked by equipment, coats or other objects that could interfere with access in an emergency.
- ◆ The pressure is at the recommended level. On extinguishers equipped with a gauge (such as that shown on the right), the needle should be in the green zone - not too high and not too low.
- ◆ The nozzle or other parts are not hindered in any way.
- ◆ The pin and tamper seal (if it has one) are intact.
- ◆ There are no dents, leaks, rust, chemical deposits and/or other signs of abuse/wear.
- ◆ Wipe off any corrosive chemicals, oil, gunk etc. that may have deposited on the extinguisher.

2. DON'T FIGHT A FIRE UNLESS:

- ◆ You call the fire department first. An extinguisher is no substitute for the fire department.



- ◆ The fire is small and not spreading. A fire can double in size every two or three minutes.

- ◆ You can get out fast if you can't control the fire. Don't get trapped. Make sure the fire is not between you and your exit.





- ◆ You have the right extinguisher for what's burning. **READ THE LABEL!** Know what types of extinguisher you have before there is a fire.

- ◆ Your extinguisher works. Inspect extinguishers once a month for signs of damage, corrosion, tampering and leaks. A partially discharged extinguisher is an empty one.

Check the gauge to insure the extinguisher is full



- ◆ You know how to use your extinguisher. It's too late to read the instructions when the fire is burning. Attend an extinguisher training class at your local fire station.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

Perform basic fire fighting

SAQA US ID		UNIT STANDARD TITLE	
12484		Perform basic fire fighting	
ORIGINATOR		REGISTERING PROVIDER	
SGB Manufacturing and Assembly Processes			
FIELD		SUBFIELD	
Field 06 - Manufacturing, Engineering and Technology		Manufacturing and Assembly	
ABET BAND	UNIT STANDARD TYPE	NQF LEVEL	CREDITS
Undefined	Regular	Level 2	4
REGISTRATION STATUS	REGISTRATION START DATE	REGISTRATION END DATE	SAQA DECISION NUMBER
Reregistered	2007-05-15	2010-05-15	SAQA 0160/05
LAST DATE FOR ENROLMENT		LAST DATE FOR ACHIEVEMENT	
2011-05-15		2014-05-15	

PURPOSE OF THE UNIT STANDARD

The skills, values and knowledge reflected in this unit standard are required by people in the field of manufacturing and engineering.

The learning outcomes in this unit standard also contribute to the exit level outcomes required for various manufacturing and engineering qualifications.

Qualifying learners can select and use the appropriate fire fighting equipment to extinguish or control fires in the workplace.

LEARNING ASSUMED TO BE IN PLACE AND RECOGNITION OF PRIOR LEARNING

This unit standard has been designed as part of a progression. It is one of a series of unit standards for safety, health and environmental protection in the field of manufacturing and engineering.

UNIT STANDARD RANGE

The scope and level of this unit standard is determined by the ranges as indicated under the specific outcomes.

UNIT STANDARD OUTCOME HEADER

N/A

Specific Outcomes and Assessment Criteria:**SPECIFIC OUTCOME 1**

Discuss and explain procedures for dealing with fires in the workplace

ASSESSMENT CRITERIA

ASSESSMENT CRITERIA

ASSESSMENT CRITERION 1

1. An understanding of procedures for dealing with fires in the workplace is demonstrated.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
 2. Appropriate fire fighting and safety equipment is selected and checked
 3. Hazards are isolated
 4. Fire is approached according to procedures
 5. Changes in the fire are monitored and responded to
 6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire
- Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures from fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 2

2. Fires are contained and/or extinguished.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
 2. Appropriate fire fighting and safety equipment is selected and checked
 3. Hazards are isolated
 4. Fire is approached according to procedures
 5. Changes in the fire are monitored and responded to
 6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire
- Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures from fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 3

3. The fire and/or the site are handed over to the appropriate personnel.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
2. Appropriate fire fighting and safety equipment is selected and checked
3. Hazards are isolated
4. Fire is approached according to procedures
5. Changes in the fire are monitored and responded to
6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire

Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures from fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 4

4. Reports on status of fire and equipment are completed.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
2. Appropriate fire fighting and safety equipment is selected and checked
3. Hazards are isolated
4. Fire is approached according to procedures
5. Changes in the fire are monitored and responded to
6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire

Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures from fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

SPECIFIC OUTCOME 2

Identify the type of fire, its context and select the appropriate fire fighting procedure

OUTCOME RANGE

Types of fires include organic materials/fibre materials, flammable liquids and gases, live electrical equipment, combustible metal fires.

Fire contexts include the nature of the area where the fire is, adjacent hazards, structural materials.

Methods of fire transmission include conduction, induction, radiation.

Combustible hazards include combustible materials and temperatures, compressed containers, electrical equipment, structure and layout of site.

ASSESSMENT CRITERIA

ASSESSMENT CRITERION 1

1. An understanding of procedures for dealing with fires in the workplace is demonstrated.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
2. Appropriate fire fighting and safety equipment is selected and checked
3. Hazards are isolated
4. Fire is approached according to procedures
5. Changes in the fire are monitored and responded to

6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire
Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:

- Applicable chemistry of fires
- Methods of fire transmission
- Types of fires
- Selection of fire fighting procedures for particular fires
- Prevention of fires
- Types, uses and operation of fire fighting and safety equipment
- Retreat procedures form fire
- Issues to consider when monitoring fire and site
- Handover procedures
- Reporting and recording requirements

2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 2

2. Fires are contained and/or extinguished.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
 2. Appropriate fire fighting and safety equipment is selected and checked
 3. Hazards are isolated
 4. Fire is approached according to procedures
 5. Changes in the fire are monitored and responded to
 6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire
- Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:

- Applicable chemistry of fires
- Methods of fire transmission
- Types of fires
- Selection of fire fighting procedures for particular fires
- Prevention of fires
- Types, uses and operation of fire fighting and safety equipment
- Retreat procedures form fire
- Issues to consider when monitoring fire and site
- Handover procedures
- Reporting and recording requirements

2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 3

3. The fire and/or the site are handed over to the appropriate personnel.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
 2. Appropriate fire fighting and safety equipment is selected and checked
 3. Hazards are isolated
 4. Fire is approached according to procedures
 5. Changes in the fire are monitored and responded to
 6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire
- Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:

- Applicable chemistry of fires
- Methods of fire transmission
- Types of fires
- Selection of fire fighting procedures for particular fires
- Prevention of fires
- Types, uses and operation of fire fighting and safety equipment
- Retreat procedures form fire
- Issues to consider when monitoring fire and site
- Handover procedures
- Reporting and recording requirements

2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 4

4. Reports on status of fire and equipment are completed.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
2. Appropriate fire fighting and safety equipment is selected and checked
3. Hazards are isolated
4. Fire is approached according to procedures
5. Changes in the fire are monitored and responded to
6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire

Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:

- Applicable chemistry of fires
- Methods of fire transmission
- Types of fires
- Selection of fire fighting procedures for particular fires
- Prevention of fires
- Types, uses and operation of fire fighting and safety equipment
- Retreat procedures from fire
- Issues to consider when monitoring fire and site
- Handover procedures
- Reporting and recording requirements

2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

SPECIFIC OUTCOME 3

Identify, select and check appropriate fire fighting and safety equipment

OUTCOME RANGE

Fire fighting equipment includes hoses, extinguishers and buckets.

Safety equipment includes breathing apparatus, fire blankets, protective clothing for fire fights, heat shields.

ASSESSMENT CRITERIA

ASSESSMENT CRITERION 1

1. An understanding of procedures for dealing with fires in the workplace is demonstrated.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
2. Appropriate fire fighting and safety equipment is selected and checked
3. Hazards are isolated
4. Fire is approached according to procedures
5. Changes in the fire are monitored and responded to
6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire

Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:

- Applicable chemistry of fires
- Methods of fire transmission
- Types of fires
- Selection of fire fighting procedures for particular fires
- Prevention of fires
- Types, uses and operation of fire fighting and safety equipment
- Retreat procedures from fire
- Issues to consider when monitoring fire and site
- Handover procedures
- Reporting and recording requirements

2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 2

2. Fires are contained and/or extinguished.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
 2. Appropriate fire fighting and safety equipment is selected and checked
 3. Hazards are isolated
 4. Fire is approached according to procedures
 5. Changes in the fire are monitored and responded to
 6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire
- Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures from fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 3

3. The fire and/or the site are handed over to the appropriate personnel.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
 2. Appropriate fire fighting and safety equipment is selected and checked
 3. Hazards are isolated
 4. Fire is approached according to procedures
 5. Changes in the fire are monitored and responded to
 6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire
- Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures from fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 4

4. Reports on status of fire and equipment are completed.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
 2. Appropriate fire fighting and safety equipment is selected and checked
 3. Hazards are isolated
 4. Fire is approached according to procedures
 5. Changes in the fire are monitored and responded to
 6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire
- Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures from fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

SPECIFIC OUTCOME 4

Fight containable/extinguishable fires

OUTCOME NOTES

- Monitor and respond to changes in the fire.

OUTCOME RANGE

Changes in a fire include changes in wind direction, backdraft, ventilation, release of hazardous substances, radiation, or as a result of incorrect application of fire fighting procedures.

Fire monitoring issues include status of fire, status of fire fighting equipment, controlling access, structural damage and equipment retrieval, fire fighting casualties.

ASSESSMENT CRITERIA

ASSESSMENT CRITERION 1

1. An understanding of procedures for dealing with fires in the workplace is demonstrated.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
2. Appropriate fire fighting and safety equipment is selected and checked
3. Hazards are isolated
4. Fire is approached according to procedures
5. Changes in the fire are monitored and responded to
6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire

Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures from fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 2

2. Fires are contained and/or extinguished.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
2. Appropriate fire fighting and safety equipment is selected and checked
3. Hazards are isolated
4. Fire is approached according to procedures
5. Changes in the fire are monitored and responded to

6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire
Understanding confirmed
- Respond to 'what if' and 'why' questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures form fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
 - Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 3

- The fire and/or the site are handed over to the appropriate personnel.

ASSESSMENT CRITERION NOTES

Indicators

- The nature of the fire is correctly identified and its context correctly assessed
- Appropriate fire fighting and safety equipment is selected and checked
- Hazards are isolated
- Fire is approached according to procedures
- Changes in the fire are monitored and responded to
- Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire
Understanding confirmed
 - Respond to 'what if' and 'why' questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures form fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
 - Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 4

- Reports on status of fire and equipment are completed.

ASSESSMENT CRITERION NOTES

Indicators

- The nature of the fire is correctly identified and its context correctly assessed
- Appropriate fire fighting and safety equipment is selected and checked
- Hazards are isolated
- Fire is approached according to procedures
- Changes in the fire are monitored and responded to
- Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire
Understanding confirmed
 - Respond to 'what if' and 'why' questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures form fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements

2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

SPECIFIC OUTCOME 5

Retreat from fire site and hand over to appropriate personnel

ASSESSMENT CRITERIA

ASSESSMENT CRITERION 1

1. An understanding of procedures for dealing with fires in the workplace is demonstrated.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
2. Appropriate fire fighting and safety equipment is selected and checked
3. Hazards are isolated
4. Fire is approached according to procedures
5. Changes in the fire are monitored and responded to
6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire

Understanding confirmed

1. Respond to `what if` and `why` questions covering:
 - < Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures from fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 2

2. Fires are contained and/or extinguished.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
2. Appropriate fire fighting and safety equipment is selected and checked
3. Hazards are isolated
4. Fire is approached according to procedures
5. Changes in the fire are monitored and responded to
6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire

Understanding confirmed

1. Respond to `what if` and `why` questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures from fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 3

3. The fire and/or the site are handed over to the appropriate personnel.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
2. Appropriate fire fighting and safety equipment is selected and checked
3. Hazards are isolated
4. Fire is approached according to procedures
5. Changes in the fire are monitored and responded to
6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire

Understanding confirmed

1. Respond to `what if` and `why` questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures form fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

ASSESSMENT CRITERION 4

4. Reports on status of fire and equipment are completed.

ASSESSMENT CRITERION NOTES**Indicators**

1. The nature of the fire is correctly identified and its context correctly assessed
 2. Appropriate fire fighting and safety equipment is selected and checked
 3. Hazards are isolated
 4. Fire is approached according to procedures
 5. Changes in the fire are monitored and responded to
 6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire
- Understanding confirmed**
1. Respond to `what if` and `why` questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures form fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
 2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task

SPECIFIC OUTCOME 6

Report/record status of fire and equipment

ASSESSMENT CRITERIA**ASSESSMENT CRITERION 1**

1. An understanding of procedures for dealing with fires in the workplace is demonstrated.

ASSESSMENT CRITERION NOTES**Indicators**

1. The nature of the fire is correctly identified and its context correctly assessed
2. Appropriate fire fighting and safety equipment is selected and checked
3. Hazards are isolated
4. Fire is approached according to procedures
5. Changes in the fire are monitored and responded to
6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire

Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures from fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task.

ASSESSMENT CRITERION 2

2. Fires are contained and/or extinguished.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
2. Appropriate fire fighting and safety equipment is selected and checked
3. Hazards are isolated
4. Fire is approached according to procedures
5. Changes in the fire are monitored and responded to
6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire

Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures from fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task.

ASSESSMENT CRITERION 3

3. The fire and/or the site are handed over to the appropriate personnel.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
2. Appropriate fire fighting and safety equipment is selected and checked
3. Hazards are isolated
4. Fire is approached according to procedures
5. Changes in the fire are monitored and responded to
6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire

Understanding confirmed

1. Respond to 'what if' and 'why' questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures from fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
2. Explain the implications of not adhering to sequence of activities and operations as described in the

specific outcomes and making decisions inappropriate to the task.

ASSESSMENT CRITERION 4

4. Reports on status of fire and equipment are completed.

ASSESSMENT CRITERION NOTES

Indicators

1. The nature of the fire is correctly identified and its context correctly assessed
 2. Appropriate fire fighting and safety equipment is selected and checked
 3. Hazards are isolated
 4. Fire is approached according to procedures
 5. Changes in the fire are monitored and responded to
 6. Retreat from the fire site is undertaken according to appropriate retreat procedures for the type of fire
- Understanding confirmed
1. Respond to `what if` and `why` questions covering:
 - Applicable chemistry of fires
 - Methods of fire transmission
 - Types of fires
 - Selection of fire fighting procedures for particular fires
 - Prevention of fires
 - Types, uses and operation of fire fighting and safety equipment
 - Retreat procedures from fire
 - Issues to consider when monitoring fire and site
 - Handover procedures
 - Reporting and recording requirements
 2. Explain the implications of not adhering to sequence of activities and operations as described in the specific outcomes and making decisions inappropriate to the task.

UNIT STANDARD ACCREDITATION AND MODERATION OPTIONS

The assessment will be governed by the policies and guidelines of the relevant Education and Training Quality Assurance Body (ETQA) who has jurisdiction over this field of learning.

The assessor will be accredited and have a technical qualification in this learning area.

The learner can be assessed in the language of his/her choice although if s/he has to report incidents or conditions to someone else, s/he will be assessed on his/her ability to report in the language commonly used in the working environment.

The learner will be assessed in the workplace or by simulation, but can submit documents, projects, test results and assignments that are not produced in the workplace or by the RPL process.

The learner can be assessed against this unit standard to obtain credits or as part of an integrated assessment for a qualification.

UNIT STANDARD ESSENTIAL EMBEDDED KNOWLEDGE

1. Names & functions of:
 - Fire fighting equipment
 - Symbols on fire fighting equipment
 - Safety equipment
2. Purpose of:
 - Identifying the nature of a fire
3. Attributes, descriptions, characteristics & properties:
 - Ideal condition of fire fighting equipment
 - Factors that change the nature of a fire
4. Cause and effect, implications of:
 - Implications of using defective fire fighting equipment
 - Implications of failing to correctly detect the nature and context of a fire
5. Procedures and techniques:
 - Fire fighting methods
 - Procedures for monitoring a fire
 - Retreat procedures for all kinds of fires
 - Handover procedures
 - Recording and reporting procedures

6. Regulations, legislation, agreements, policies, standards:
 - Applicable safety, health and environmental protection legislation
 - Applicable organisational policies and procedures
7. Theory: rules, principles, laws:
 - Applicable chemistry of fire: the triangle of combustion, elimination factors, transmission or spread
8. Relationships, systems:
 - Relationship between the nature and context of a fire and the fire fighting technique applied

UNIT STANDARD DEVELOPMENTAL OUTCOME

N/A

UNIT STANDARD LINKAGES

N/A

Critical Cross-field Outcomes (CCFO):

UNIT STANDARD CCFO IDENTIFYING

Identify and solve problems:

- Related to identifying the nature and context of a fire

UNIT STANDARD CCFO WORKING

Work effectively with others:

- To fight fires

UNIT STANDARD CCFO ORGANISING

Organise and manage myself and my activities:

- To ensure that all required fire fighting equipment is at hand when needed

UNIT STANDARD CCFO COLLECTING

Collect, analyse, organise and critically evaluate information:

- Related to compiling a report on the status of the fire and fire fighting equipment

UNIT STANDARD CCFO COMMUNICATING

Communicate effectively:

- When handing over the fire and/or site to emergency personnel

UNIT STANDARD CCFO SCIENCE

Use science and technology effectively and critically:

- Apply chemistry of fire concepts

UNIT STANDARD CCFO DEMONSTRATING

Demonstrate an understanding of the world as a set of related systems:

- Explain the relationship between the nature and context of a fire and the fire fighting technique applied

UNIT STANDARD ASSESSOR CRITERIA

N/A

UNIT STANDARD NOTES

This Unit Standard was originally registered on 3 December 2003 under SAQA decision number 0151/03. It has been re-registered on 11 February 2004 under SAQA 2552/04.